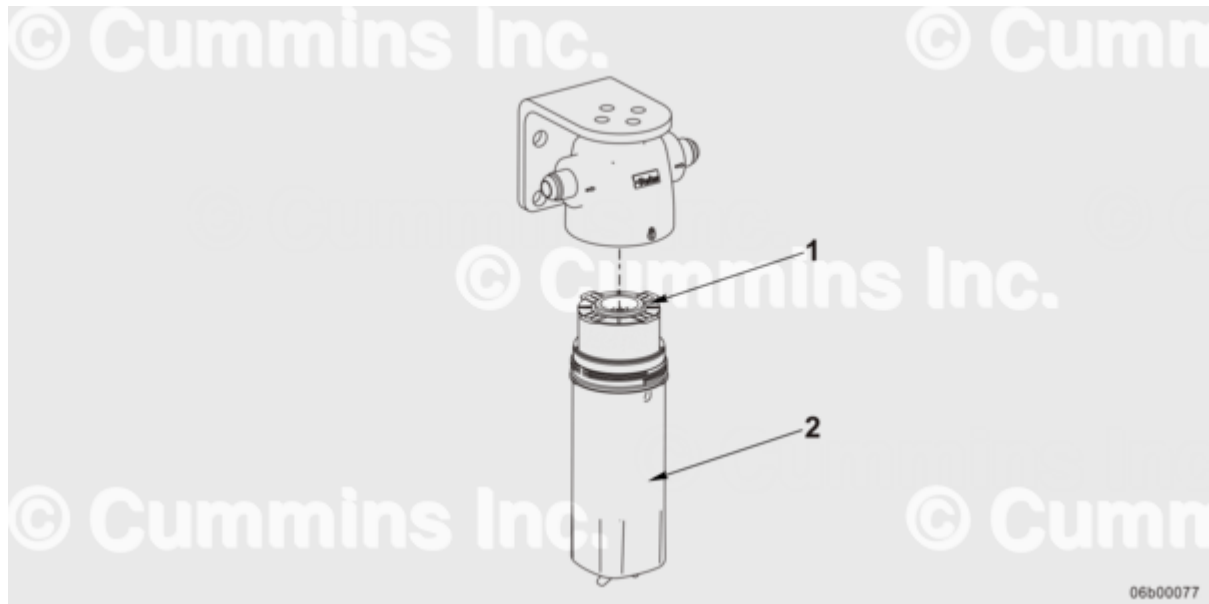


Trainings ▾

Exploded View



Fuel Filter (Canister Type) Exploded View

1. Fuel filter element - quantity 1
2. Fuel filter canister - quantity 1

Select Service Tools

Recommended Cummins® Service Tools

- Oil filter wrench, Part Number 3400158
- Combustible gas detector, Part Number 3165179

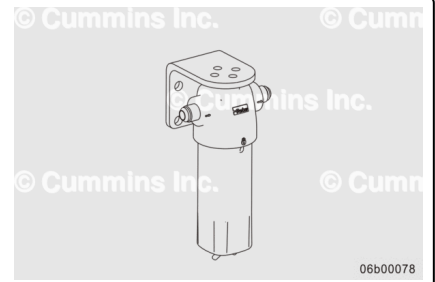
Additional Service Items

- No additional service items required.

General Information

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



Filtration and separation of oil from the fuel is important for trouble-free operation and long life of the fuel system. In certain cases, oil is introduced into the fuel from gas compressors when the fuel is pressurized. Some of the components in the fuel system are susceptible to oil contamination.

Each fuel filter element has a valve that **must** be opened daily to drain the collected oil. Regular maintenance, including draining oil from the filter, is essential to keep the oil out of the fuel system.

Note : For Compressed Natural Gas (CNG) engine, the primary fuel filter is supported by Original Equipment Manufacturer (OEM) and the secondary fuel filter is identical to the fuel filter of Liquefied Natural Gas (LNG) engine.

Preparatory Steps

⚠ WARNING ⚠

Natural gas is explosive and flammable. Always be sure to maintain adequate ventilation in the work area. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas with shared ventilation to reduce the possibility of severe personal injury or death when working on a natural gas system.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Close the main vehicle fuel supply valve.
- Start the engine and allow it to run at idle.
- Allow the engine to run until it shuts down.
- Disconnect the batteries. See the equipment manufacturer service information.

Drain

Use your hand to open the drain valve. Turn the valve **clockwise** approximately 1-½ to 2 turns until draining occurs.

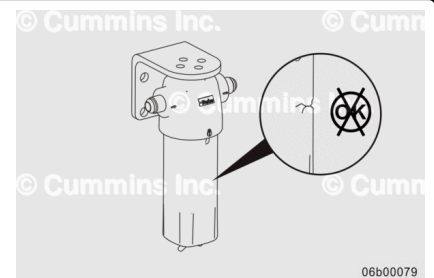
Drain the oil from the fuel filter.

When closing the drain valve, do **not** overtighten the valve. Overtightening can damage the threads. Turn the valve **counterclockwise** to close the drain valve.

Inspect for Reuse

Inspect the fuel filter canister for cracks, distortion, or other damage.

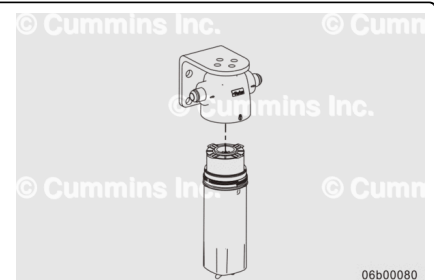
Replace the support bracket if damage is found.



Remove

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



Clean the area around the fuel filter area.

Use the oil filter wrench, Part Number 3400158, to remove the fuel filter from the fuel filter head.

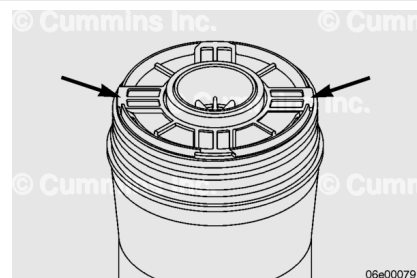
Remove and discard the filter element.

Remove and discard the o-ring seal on the filter canister.



Install a new filter element.

Make sure the tabs of the filter element are properly seated in the canister.



Install

Use the following procedure for correct fuel filter. Refer to Procedure 018-024 in Section V. (</qs3/pubsys2/xml/en/procedures/1016/1016-018-024.html>)

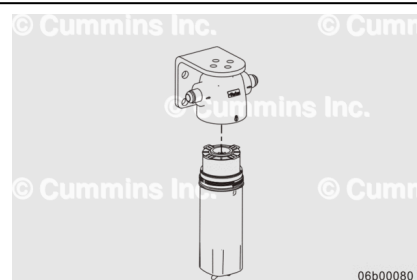
⚠ CAUTION ⚠

Mechanical overtightening can distort the threads as well as damage the filter element seal or filter can.

Install the filter on the filter head.

Install to the point of first contact for the filter and head.

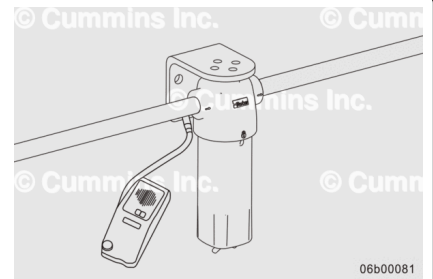
Tighten the filter by hand an additional $\frac{1}{2}$ to $\frac{3}{4}$ turn after the gasket contacts the fuel filter head surface, or as specified by the fuel filter manufacturer.



Finishing Steps

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. See equipment manufacturer service information.
- Open the vehicle main fuel supply valve.
- Start the engine and let it idle for 1 minute.
- Turn key switch OFF.
- Turn key switch ON. Do **not** engage the starting motor. Check all connections for gas leaks with a combustible gas detector, Part Number 3165179.
- If leaks are found, close the valve, turn keyswitch OFF, and repair leaks immediately.